



AX7010 Development Board User Manual

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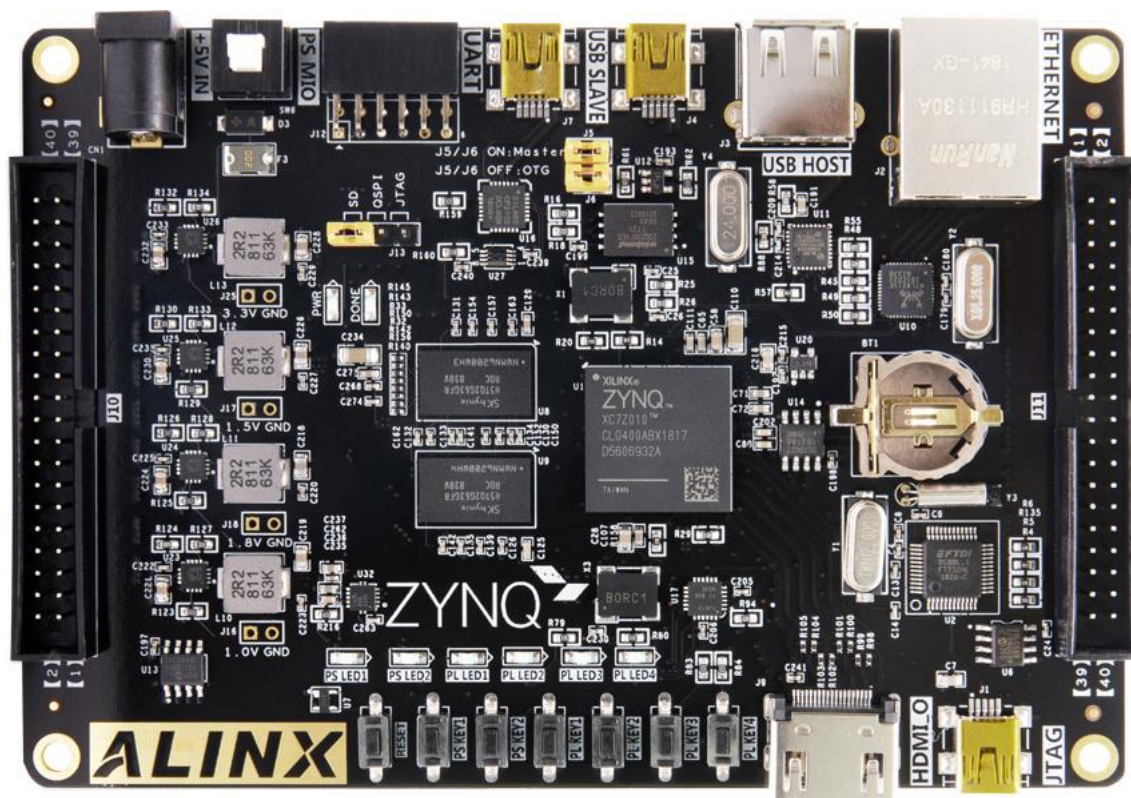
Document revision history

chapter	Revision Overview
V1.0	
all	First version released
V2.0	
/	1、Corrected pin assignment errors in J11 2、The QSPI capacity error description has been corrected. 3、PCB modification, EEPROM SDA pin modification
V3.0	
6.7 PS PMOD Connector 6.2 DDR 3 DRAM	Corrected PMOD pin assignment definition error Modify DDR3 model

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The 2016 development board based on the Xilinx ZYNQ7000 development platform, model number AX7010, has been officially released. This development platform is a solution based on the Xilinx Zynq7000 SOC chip. It integrates a dual-core ARM Cortex-A9 processor and FPGA programmable logic onto a single chip using ARM+FPGA SOC technology. It uses the Xilinx Zynq7000 series XC7Z010-1CLG400C as its core processor, and features abundant hardware resources and peripheral interfaces on both the ARM and FPGA sides. Adhering to the design philosophy of "refined, practical, and simple," it is suitable not only for early software verification by software engineers but also for hardware design by hardware developers, i.e., system collaboration between software and hardware, accelerating project development.



01 Introduction

Here is a brief introduction to the features of the ZYNQ7000 development platform AX7010.

This development board uses Xilinx's Zynq 7000 series chip , model XC7Z010-1CLG400C , in a 400-pin FBGA package. The Zynq 7000 chip can be divided into a Processor System (PS) and a Programmable Logic (PL) section. On the AX7010 development board, both the PS and PL sections of the Zynq 7000 are equipped with abundant external interfaces and devices, facilitating user operation and functional verification. Additionally, the development board integrates a Xilinx USB Cable programmer circuit, allowing users to download and debug code using only a USB cable. Figure 1.1.1 shows a schematic diagram of the entire AX7010 system.

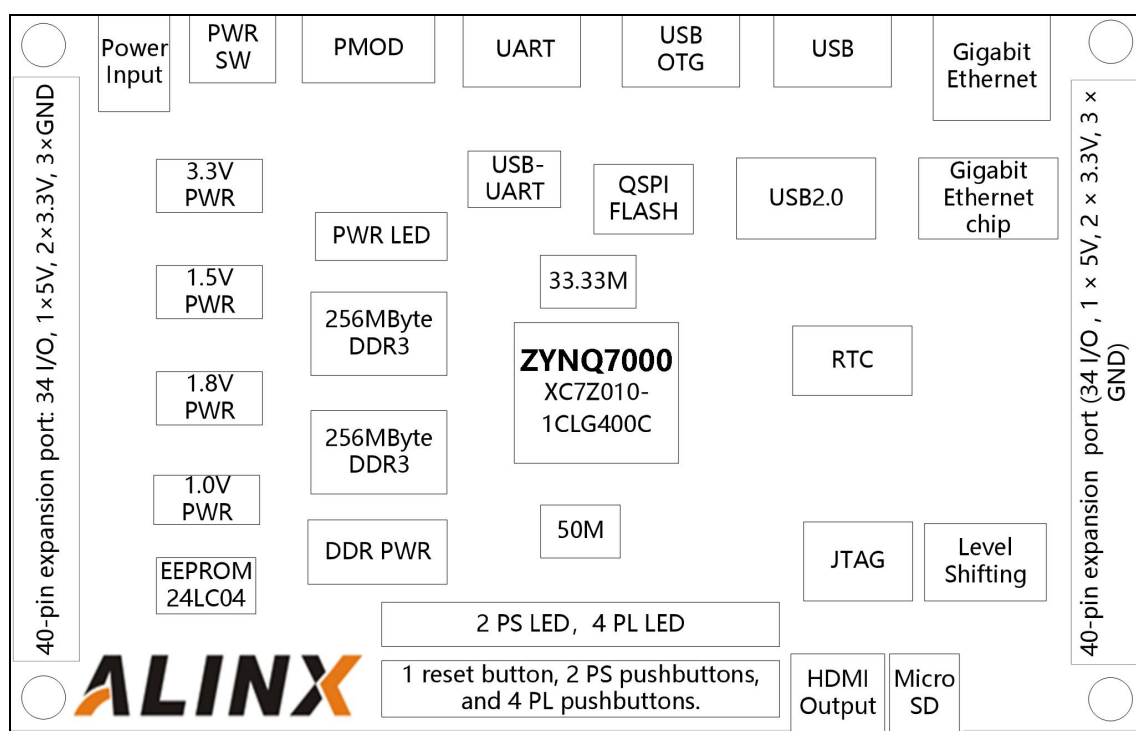


Figure 1.1.1 - Schematic diagram of AX7010 structure

This diagram shows the interfaces and functions that our development platform can contain.

- +5V power input, maximum 2A current protection;
- Xilinx ARM+FPGA chip Zynq-7000 XC7Z010-1CLG400C
- Two high-capacity 2Gbit (total 4Gbit) high-speed DDR3 SDRAM chips can be used as cache for ZYNQ chip data or as memory for the operating system.
- A 256Mbit QSPI FLASH chip can be used as storage for system files and user data of the ZYNQ chip;
- One 10/100M /1000M Ethernet RJ-45 interface, which can be used for Ethernet data exchange with computers or other network devices;
- One HDMI input/output interface enables 1080P video transmission;

- One high-speed USB 2.0 HOST interface can be used to connect USB peripherals such as mice, keyboards, and USB flash drives to the development board;
- One high-speed USB 2.0 OTG interface for OTG communication with PCs or USB devices;
- One USB UART interface for serial communication with a PC or external devices;
- One RTC real-time clock, with a battery holder; the battery model is CR1220.
- One IIC interface EEPROM 24LC04;
- Six user-defined light-emitting diodes (LEDs), controlled by two PS (Power Sealant) and four PL (Power Probe) controllers;
- 7 buttons: 1 CPU reset button, 2 PS control buttons, and 4 PL control buttons;
- The board has an onboard 33.333 MHz active crystal oscillator to provide a stable clock source for the PS system, and a 50 MHz active crystal oscillator to provide an additional clock for the PL logic.
- Two 40-pin expansion ports (2.54mm pitch) are used to expand the I/O of the ZYNQ's PL section. They can connect to expansion modules such as 7-inch TFT modules, camera modules, and AD/DA modules.
- A 12-pin expansion port (2.54mm pitch) is used to expand the MIO of ZYNQ's PS system;
- One USB JTAG port allows for debugging and downloading of the ZYNQ system via a USB cable and onboard JTAG circuitry ;
- A single Micro SD card slot (on the back of the development board) is used to store the operating system image and file system.

02 Structural Dimensions

The development board measures a compact 130mm x 90mm and features an 8-layer PCB design. Four screw holes with a diameter of 3.5mm are located around the board for securing it. A DXF structural diagram is provided in the documentation.

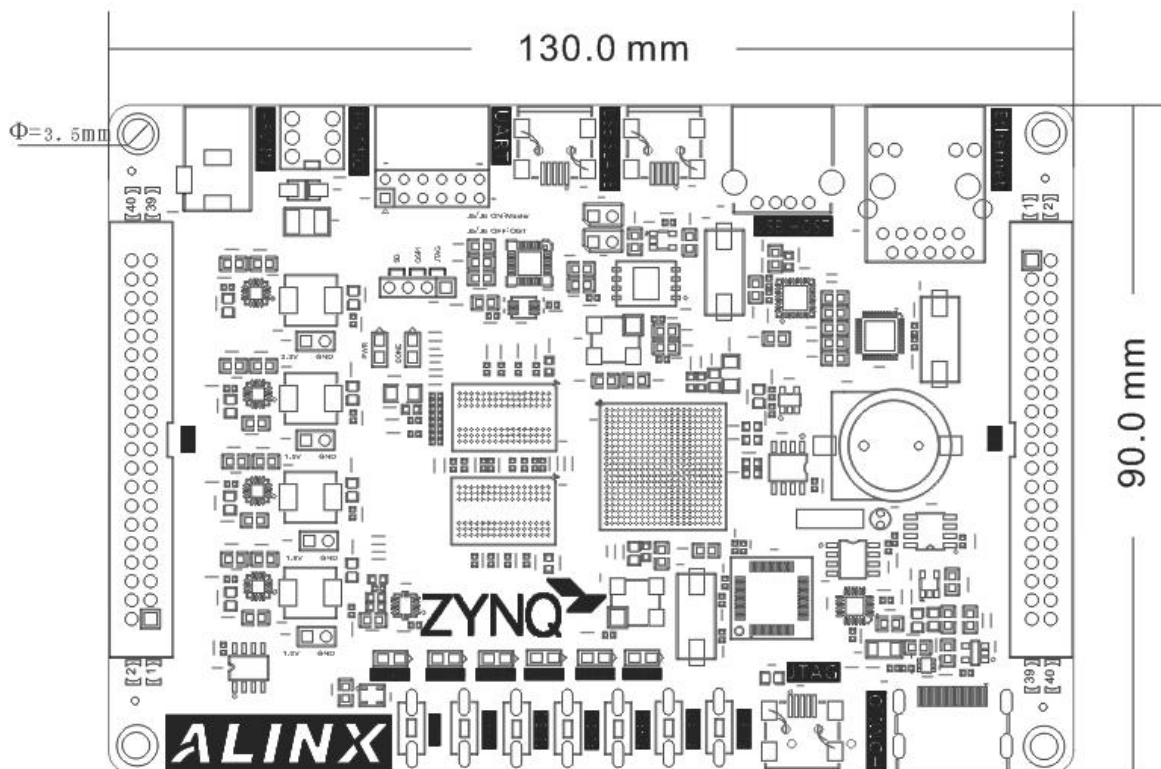


Figure 2.1.1 - Structural Diagram

03 Power Supply

Power Input: The development board requires a DC 5V power supply. Please use the power supply that came with the development board. Do not use other power supplies to avoid damaging the development board. The power supply design diagram on the development board is shown below:

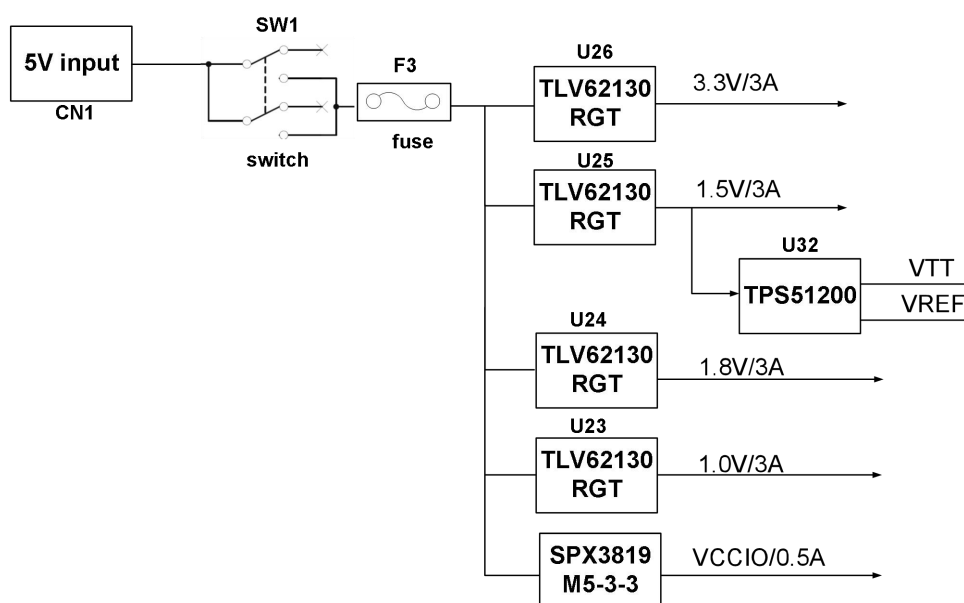


Figure 3.1.1 - Power interface section in the schematic diagram

The development board is powered by +5V, which is converted into four power supplies (+3.3V, +1.5V, +1.8V, and +1.0V) by a four-channel DC/DC power supply chip TLV62130RGT , with each channel capable of outputting up to 3A. A single LDO SPX3819M5-3-3 generates the VCCIO power supply. The VCCIO is primarily designed to power ZYNQ's BANK35 ; by replacing the LDO chip, the BANK35's I/O can be adapted to different voltage standards. The 1.5V is generated by TI's TPS51200 to provide the VTT and VREF voltages required for DDR3. The functions of each power supply are shown in the table below.

Power supply	Function
+3.3V	ZYNQ VCCIO, Ethernet, Serial, HDMI, RTC, FLASH, EEPROM, and SD card
+1.8V	ZYNQ auxiliary voltage, ZYNQ PLL, ZYNQ Bank501 VCCIO, Ethernet, USB 2.0
+1.0V	ZYNQ, core voltage
+1.5V	DDR3, ZYNQ Bank502
VREF, VTT	DDR3
VCCIO	ZYNQ Bank35

Table 3.1.1 - Power Function Allocation

Because the power supply of the PS and PL sections of the ZYNQ has a power-on sequence requirement, we have designed the circuit according to the ZYNQ power supply requirements, with the power-on sequence being 1.0V -> 1.8V -> 1.5V -> 3.3V -> VCCIO .

When designing the PCB, we used an 8-layer PCB with dedicated power and GND layers, ensuring excellent power stability for the entire development board. We also included test points for each power supply on the PCB to allow users to verify the voltage on the board.

04 ZYNQ7000

The development board uses a Xilinx Zynq 7000 series chip , model XC7Z010-1CLG400C . The chip's System-on-PS (PS) integrates two ARM Cortex™-A9 processors, AMBA® interconnect, internal memory, external memory interface, and peripherals. These peripherals mainly include a USB bus interface, Ethernet interface, SD/SDIO interface, I2C bus interface, CAN bus interface, UART interface, GPIO, etc. The PS can operate independently and boot upon power-on or reset. The overall block diagram of the ZYNQ7000 chip is shown in Figure 4.1.1.

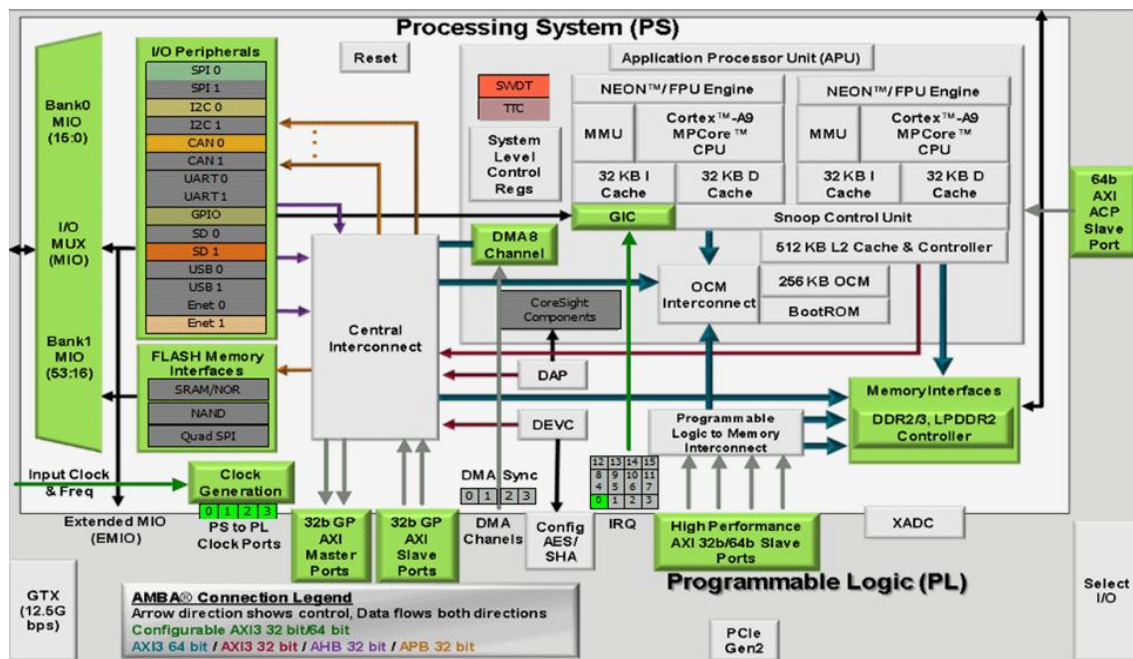


Figure 4.1.1 - Overall block diagram of the ZYNQ 7000 chip

The main parameters of the PS system are as follows:

- Application processor based on ARM dual-core Cortex A9
- Each CPU has 32KB of L1 instruction and data cache, and 512KB of L2 cache shared by both CPUs.
- On-chip boot ROM and 256KB on-chip RAM
- External storage interface, supports 16/32 bit DDR2 and DDR3 interfaces.
- Two Gigabit network cards support: Divergent-Convergent DMA, GMII, RGMII, and SGMII interfaces.
- Two USB 2.0 OTG ports, each supporting up to 12 nodes.
- Two CAN2.0B bus interfaces
- Two SD cards, SDIO, MMC compatible controller
- 2 SPI interfaces, 2 UARTs, 2 I2C interfaces
- 4 groups of 32-bit GPIOs, 54 (32+22) as PS system I/O, 64 connected to PL
- High-bandwidth connections within PS and from PS to PL

The main parameters of the PL logic section are as follows:

- Logic Cells: 28K;
- Lookup table LUTs: 17600
- Flip-flops : 35200
- Multiplier 18x25 MACCs: 80;

- Block RAM: 2.1Mb;
- Two analog-to-digital converters, capable of measuring on-chip voltage, temperature sensing, and up to 17 external differential input channels, 1 Mbps.

XC7Z010-1CLG400C chip is a BGA package with 400 pins and a pin pitch of 0.8mm. To reiterate, when using a BGA-packaged chip, the pin names become letter-and-number combinations, such as E3, G3, etc. Therefore, when viewing a schematic, the letter-and-number combination represents the BGA pins.

4.1 JTAG Interface

First, let's discuss the JTAG debugging interface of the AX7010 development board. The JTAG download and debugging circuitry is integrated on the board, so users don't need to purchase an additional Xilinx downloader. A single USB cable is sufficient for ZYNQ development and debugging. The AX7010 development board uses an FTDI USB bridge chip, FT232HL, to facilitate data communication between the PC's USB port and the ZYNQ's JTAG debugging signals TCK, TDO, TMS, and TDI.

On the AX7010 development board, the JTAG interface is in the form of a USB interface. Users can connect a PC and the JTAG interface through the USB cable we provide for ZYNQ system debugging.

4.2 FPGA Power Supply System

Next, let's discuss the power supply design of the AX7010. The ZYNQ chip's power supply is divided into a PS system section and a PL logic section, and the power supplies for these two sections operate independently. Both the PS system and PL logic sections have a specific power-on sequence; an incorrect power-on sequence may cause the ARM and FPGA systems to malfunction.

The PS section has four power supply pins: VCCPINT, VCCPAUX, VCCPLL, and PS VCCO. VCCPINT is the PS core power supply pin, connected to 1.0V; VCCPAUX is the PS system auxiliary power supply pin, connected to 1.8V; VCCPLL is the power supply pin for the PS's internal clock PLL, also connected to 1.8V; PS VCCO is the bank voltage, including VCCO_MIO0, VCCO_MIO1, and VCCO_DDR. The power supply connection varies depending on the connected peripheral. On the AX7010 development board, VCC_MIO0 is connected to 3.3V, VCCO_MIO1 to 1.8V, and VCCO_DDR to 1.5V. The PS system requires the following power-on sequence: first VCCPINT, then VCCPAUX and VCCPLL, and finally PS VCCO. The power-off sequence is the reverse.

The power supplies for the PL section are VCCINT, VCCBRAM, VCCAUX, and VCCO. VCCINT is the FPGA core power supply pin, connected to 1.0V; VCCBRAM is the FPGA Block RAM power supply pin, connected to 1.0V; VCCAUX is the FPGA auxiliary power supply pin, connected to 1.8V; VCCO is the voltage for each PL bank, including BANK13, BANK34, and BANK35. On the AX7010 development board, the bank voltages are connected to 3.3V. The PL system requires the following power-on sequence: first VCCINT, then VCCBRAM, then VCCAUX, and finally VCCO. If the voltages for VCCINT and VCCBRAM are the same, they can be powered on simultaneously. The power-off sequence is the reverse.

4.3 ZYNQ Startup Configuration

The AX7010 development platform supports three boot modes: JTAG debug mode, QSPI FLASH, and SD card boot mode. After power-on, the ZYNQ 7010 chip detects the level of the MIO port to determine the boot mode. Users can select different boot modes via the jumper on the J13 pin of the core board . The J13 boot mode configuration is shown in Table 4.3.1 below .

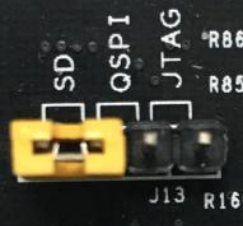
J13	Hat position	Startup mode
	Connect the two pins on the left	SD Card
	Connect the two middle pins	QSPI FLASH
	Connect the two pins on the right side	JTAG

Table 4.3.1- J13 Boot Mode Configuration

05 Clock Configuration

The AX7010 development board provides active clocks for the PS system and PL logic sections, allowing the PS system and PL logic to operate independently.

5.1 PS System Clock Source

The ZYNQ chip provides a 33.333MHz clock input to the PS section via the X1 crystal oscillator on the development board. The clock input is connected to the PS_CLK_500 pin of the ZYNQ chip's BANK500. Clock pin assignment:

Signal name	ZYNQ pin
PS_CLK_500	E7

Table 5.1.1 - PS System Clock Source

5.2 PL System Clock Source

The AX7010 development board provides a single-ended 50MHz PL system clock source, powered by 3.3V. The crystal oscillator output is connected to the FPGA's global clock (MRCC), which can be used to drive user logic circuits within the FPGA . PL clock pin assignment:

Signal name	ZYNQ pin
PL_GCLK	U18

Table 5.2.1 - PL Clock Pin Assignment

06 PS peripherals

Because the ZYNQ consists of an ARM system PS section and an FPGA logic PL section, some peripherals on the development board are connected to the PS's I/O, while others are connected to the PL's I/O. First, we will introduce the peripherals connected to the PS section.

6.1 QSPI Flash

The development board comes with a 256 MB Quad-SPI FLASH chip, model W25Q256, which uses the 3.3V CMOS voltage standard. Due to the non-volatile nature of the QSPI FLASH, it can be used as a system boot device to store the system boot image. These images mainly include the FPGA bit file, ARM application code, and other user data files. Specific models and related parameters of the QSPI FLASH are shown in Table 6.1.1 .

Position	Chip type	Capacity	Factory
U15	W25Q256	32M Byte	Winbond

Table 6.1.1 - QSPI Flash Models and Parameters

BANK500 of the ZYNQ chip . In the system design, these PS-side GPIO ports need to be configured as QSPI FLASH interfaces. Figure 6.1.1 shows a schematic diagram of the QSPI Flash hardware connection.

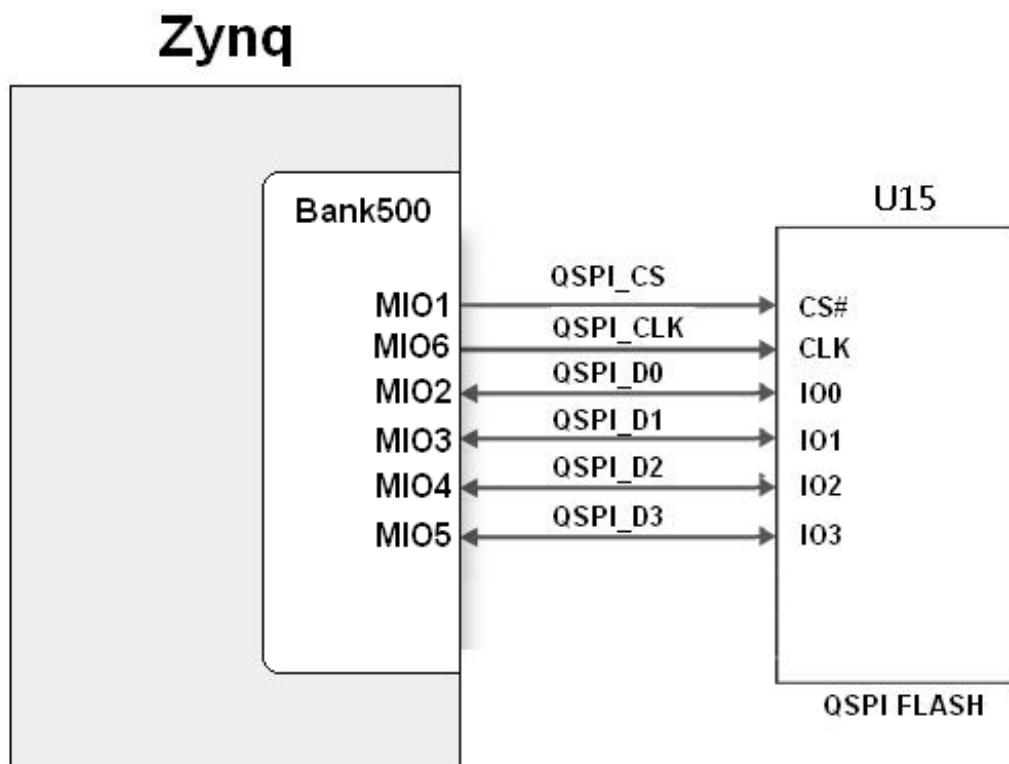


Figure 6.1.1 - QSPI Flash Connection Diagram

Configure chip pin assignments:

Signal name	ZYNQ pin name	ZYNQ pin number
QSPI_CLK	PS_MIO6_500	A5
QSPI_CS	PS_MIO1_500	A7
QSPI_D0	PS_MIO2_500	B8
QSPI_D1	PS_MIO3_500	D6
QSPI_D2	PS_MIO4_500	B7
QSPI_D3	PS_MIO5_500	A6

Table 6.1.2 - QSPI Flash Pin Assignment

6.2 DDR3 DRAM

The AX7010 development board is equipped with two SK Hynix 2Gbit (512 MB) DDR3 chips (totaling 4Gbit) , model H5TQ2G63FFR (compatible with MT41J128M16HA-125) . The DDR bus width is 32 bits. The maximum operating speed of DDR3 SDRAM is 533 MHz (data rate 1066 Mbps) . This DDR3 memory system is directly connected to the memory interface of the ZYNQ processing system (PS) BANK 502. The specific configuration of the DDR3 SDRAM is shown in Table 6.2.1 below .

Position	Chip type	Capacity	Factory
U8 , U9	H5TQ2G63FFR-RDC	128 M x 16bit	SKHynix

Table 6.2.1 - DDR3 SDRAM Configuration

The hardware design of DDR3 requires strict consideration of signal integrity. During circuit and PCB design, we have fully considered matching resistors/terminating resistors, trace impedance control, and trace length control to ensure the high-speed and stable operation of DDR3. A hardware connection diagram of DDR3 DRAM is shown in Figure 6.2.1 :

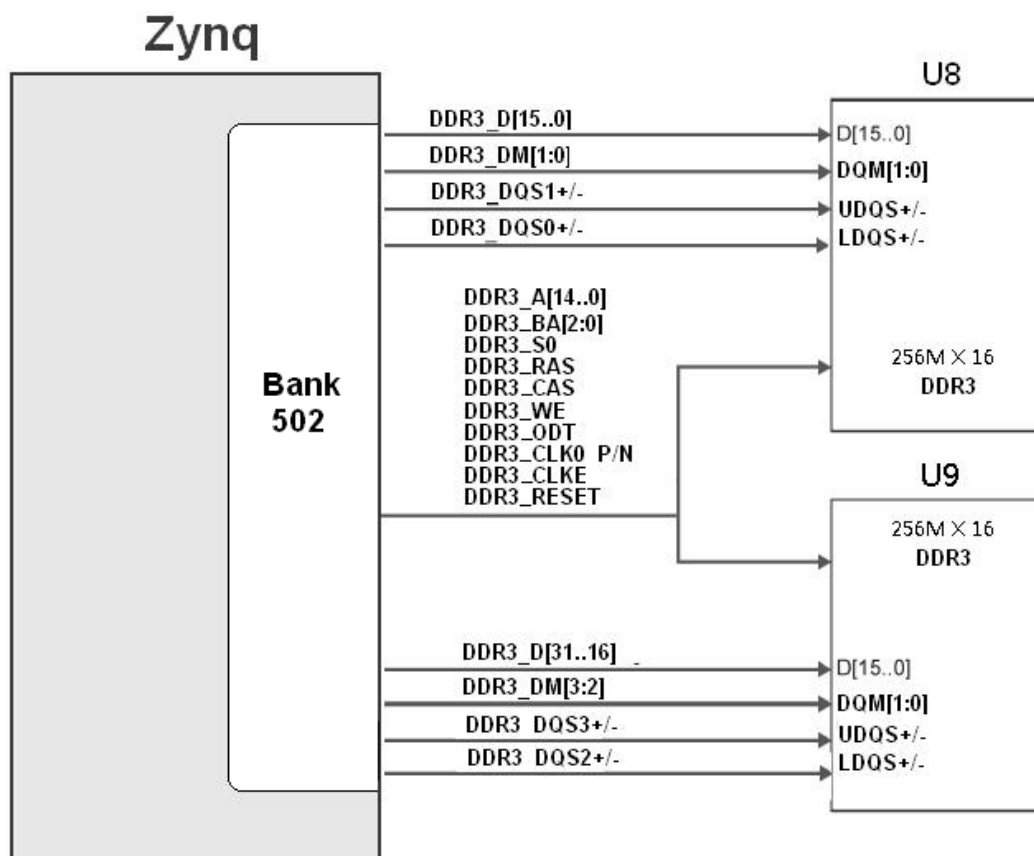


Figure 6.2.1 - Partial Schematic Diagram of DDR3 DRAM

DDR3 DRAM pin assignment:

Signal name	ZYNQ pin name	ZYNQ pin number
DDR3_DQS0_P	PS_DDR_DQS_P0_502	C2
DDR3_DQS0_N	PS_DDR_DQS_N0_502	B2
DDR3_DQS1_P	PS_DDR_DQS_P1_502	G2
DDR3_DQS1_N	PS_DDR_DQS_N1_502	F2
DDR3_DQS2_P	PS_DDR_DQS_P2_502	R2
DDR3_DQS2_N	PS_DDR_DQS_N2_502	T2
DDR3_DQS3_P	PS_DDR_DQS_P3_502	W5
DDR3_DQS 3_N	PS_DDR_DQS_N3_502	W4
DDR3_D Q[0]	PS_DDR_DQ0_502	C3
DDR3_D Q[1]	PS_DDR_DQ1_502	B3

DDR3_D Q[2]	PS_DDR_DQ2_502	A2
DDR3_D Q[3]	PS_DDR_DQ3_502	A4
DDR3_D Q[4]	PS_DDR_DQ4_502	D3
DDR3_D Q[5]	PS_DDR_DQ5_502	D1
DDR3_D Q[6]	PS_DDR_DQ6_502	C1
DDR3_D Q[7]	PS_DDR_DQ7_502	E1
DDR3_D Q[8]	PS_DDR_DQ8_502	E2
DDR3_D Q[9]	PS_DDR_DQ9_502	E3
DDR3_D Q[10]	PS_DDR_DQ10_502	G3
DDR3_D Q[11]	PS_DDR_DQ11_502	H3
DDR3_D Q[12]	PS_DDR_DQ12_502	J3
DDR3_D Q[13]	PS_DDR_DQ13_502	H2
DDR3_D Q[14]	PS_DDR_DQ14_502	H1
DDR3_D Q[15]	PS_DDR_DQ15_502	J1
DDR3_D Q[16]	PS_DDR_DQ16_502	P1
DDR3_D Q[17]	PS_DDR_DQ17_502	P3
DDR3_D Q[18]	PS_DDR_DQ18_502	R3
DDR3_D Q[19]	PS_DDR_DQ19_502	R1
DDR3_D Q[20]	PS_DDR_DQ 20_502	T4
DDR3_D Q[21]	PS_DDR_DQ 21_502	U4
DDR3_D Q[22]	PS_DDR_DQ 22_502	U2
DDR3_D Q[23]	PS_DDR_DQ 23_502	U3
DDR3_D Q[24]	PS_DDR_DQ 24_502	V1
DDR3_D Q[25]	PS_DDR_DQ 25_502	Y3

DDR3_D Q[26]	PS_DDR_DQ 26_502	W1
DDR3_D Q[27]	PS_DDR_DQ 27_502	Y4
DDR3_D Q[28]	PS_DDR_DQ 28_502	Y2
DDR3_D Q[29]	PS_DDR_DQ 29_502	W3
DDR3_D Q[30]	PS_DDR_DQ 30_502	V2
DDR3_D Q[31]	PS_DDR_DQ 31_502	V3
DDR3_DM0	PS_DDR_DM0_502	A1
DDR3_DM 1	PS_DDR_DM 1_502	F1
DDR3_DM 2	PS_DDR_DM 2_502	T1
DDR3_DM 3	PS_DDR_DM 3_502	Y1
DDR3_A[0]	PS_DDR_A0_502	N2
DDR3_A[1]	PS_DDR_A 1_502	K2
DDR3_A[2]	PS_DDR_A 2_502	M3
DDR3_A[3]	PS_DDR_A 3_502	K3
DDR3_A[4]	PS_DDR_A 4_502	M4
DDR3_A[5]	PS_DDR_A 5_502	L1
DDR3_A[6]	PS_DDR_A 6_502	L4
DDR3_A[7]	PS_DDR_A 7_502	K4
DDR3_A[8]	PS_DDR_A 8_502	K1
DDR3_A[9]	PS_DDR_A 9_502	J4
DDR3_A[10]	PS_DDR_A 10_502	F5
DDR3_A[11]	PS_DDR_A 11_502	G4
DDR3_A[12]	PS_DDR_A 12_502	E4
DDR3_A[13]	PS_DDR_A 13_502	D4

DDR3_A[14]	PS_DDR_A 14_502	F4
DDR3_BA[0]	PS_DDR_BA0_502	L5
DDR3_BA[1]	PS_DDR_BA 1_502	R4
DDR3_BA[2]	PS_DDR_BA 2_502	J5
DDR3_S0	PS_DDR_CS_B_502	N1
DDR3_RAS	PS_DDR_RAS_B_502	P4
DDR3_CAS	PS_DDR_CAS_B_502	P5
DDR3_WE	PS_DDR_WE_B_502	M5
DDR3_ODT	PS_DDR_ODT_502	N5
DDR3 RESET	PS_DDR_DRST_B_502	B4
DDR3_CLK_P	PS_DDR_CKP_502	L2
DDR3_CLK_N	PS_DDR_CKN_502	M2
DDR3_CKE	PS_DDR_CKE_502	N3

Table 6.2.2 - DDR3 DRAM Pin Assignment

6.3 Gigabit Ethernet Interface

The AX7010 development board provides network communication services through the Realtek RTL8211E-VL Ethernet PHY chip. The Ethernet PHY chip connects to the GPIO interface of the Zynq PS-side BANK501. The RTL8211E-VL chip supports 10/100/1000 Mbps network transmission rates and communicates with the MAC layer of the Zynq7000 PS system via the RGMII interface. The RTL8211E-VL supports MDI/MDX auto-adaptation, various speed auto-adaptation, Master/Slave auto-adaptation, and supports PHY register management via the MDIO bus.

Upon power-on, the RTL8211E-VL detects the level states of certain I/O pins to determine its operating mode. Table 6.3.1 describes the default settings of the GPHY chip after power-on..

Configure Pin	illustrate	Configuration value
PHYAD[2:0]	PHY address in MDIO/MDC mode	PHY Address is 001
SELRGV	RGMII 1.8V or 1.5V level selection	1.8V

AN[1:0]	Self-negotiation configuration	(10/100/1000M) Adaptive
RX Delay	RX clock 2ns delay	Delay
TX Delay	TX clock 2ns delay	Delay

Table 6.3.1 - Default Configuration Values for PHY Chip

When the network is connected to Gigabit Ethernet, the FPGA and the PHY chip RTL8211E-VL communicate via the RGMII bus for data transmission. The transmission clock is 125MHz, and the data is sampled on the rising and falling edges of the clock.

When the network is connected to a 100 Mbps Ethernet connection, data transmission between the FPGA and the PHY chip RTL8211E-VL is conducted via the RMII bus, with a transmission clock of 25 MHz. Data is sampled on both the rising and falling edges of the clock.

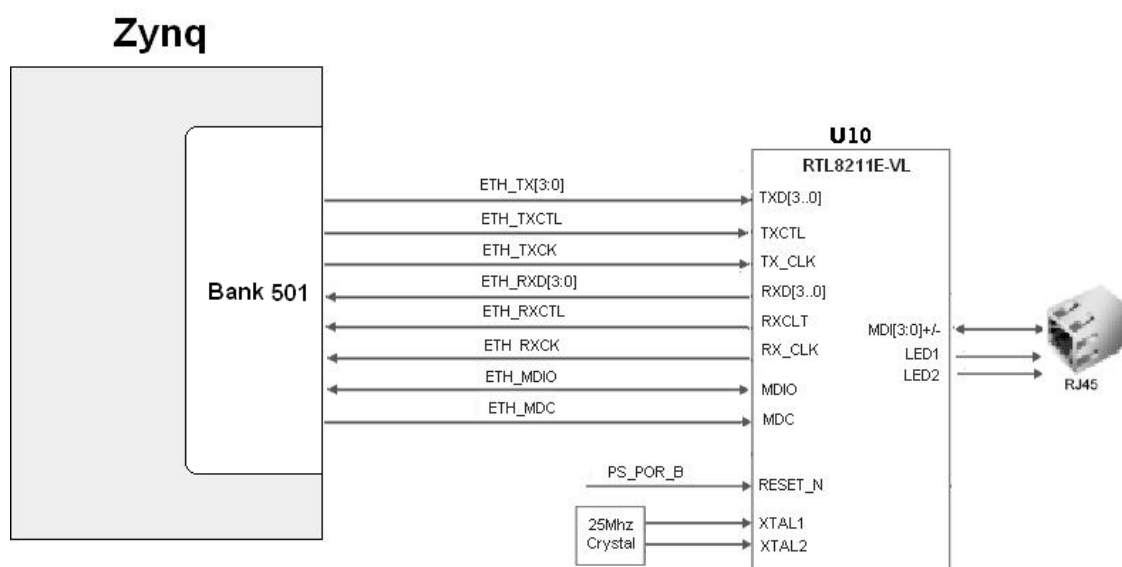


Figure 6.3.1 - Schematic diagram of FPGA and PHY connection

Ethernet pin assignments are as follows:

Signal name	ZYNQ pin name	ZYNQ pin number	Remark
ETH_GCLK	PS_MIO16_501	A19	RGMII Transmit Clock
ETH_TXD0	PS_MIO17_501	E14	Send data bit0
ETH_TXD1	PS_MIO18_501	B18	Send data bit1
ETH_TXD2	PS_MIO19_501	D10	Send data bit2
ETH_TXD3	PS_MIO20_501	A17	Send data bit 3

ETH_TXCTL	PS_MIO 21_501	F14	Send enable signal
ETH_RXCK	PS_MIO 22_501	B17	RGMII Receive Clock
ETH_RXD0	PS_MIO 23_501	D11	Receive data Bit0
ETH_RXD1	PS_MIO 24_501	A16	Receive data Bit1
ETH_RXD2	PS_MIO 25_501	F15	Receive data Bit2
ETH_RXD3	PS_MIO 26_501	A15	Receive data Bit3
ETH_RXCTL	PS_MIO 27_501	D13	Received data valid signal
ETH_MDC	PS_MIO 52_501	C10	MDIO manages the clock
ETH_MDIO	PS_MIO 53_501	C11	MDIO Management Data

Table 6.3.2 - Ethernet Pin Assignment

6.4 USB 2.0

The AX7010 uses a 1.8V, high-speed USB3320C-EZK transceiver that supports the ULPI standard interface. The ZYNQ's USB bus interface connects to the USB3320C-EZK transceiver, enabling high -speed data communication in USB 2.0 Host and Slave modes . The USB3320C 's data and control signals are connected to the BANK501 I/O port on the ZYNQ chip's PS side. A 24MHz crystal oscillator provides the system clock for the USB3320C .

The development board provides users with two USB ports : a Host USB port and a Slave USB port. These are a USB Type-A connector and a Micro USB connector, respectively . This allows users to easily connect different USB peripherals . Users can switch between Host and Slave modes using jumpers on J5 and J6 of the development board. Table 6.4.1 provides an explanation of the mode switching process.

J5, J6 status	USB mode	illustrate
Install jumper caps on J5 and J6.	HOST mode	The development board acts as the master device, while the USB port connects to peripherals such as a mouse, keyboard, and USB ports.
J5 and J6 do not require jumper caps.	OTG/Slave mode	The development board is connected to the computer via USB as a slave device.

Table 6.4.1 - USB Interface Mode Switching Instructions

A schematic diagram of the connection between the ZYNQ processor and the USB3320C-EZK chip is shown in Figure 6.4.1 :

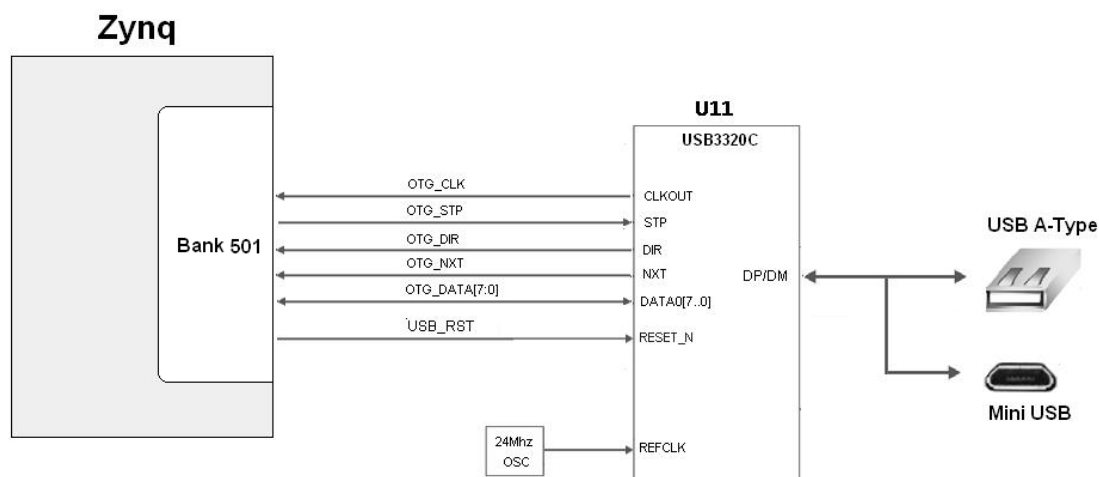


Figure 6.4.1 - Schematic diagram of the connection between Zynq7000 and USB chip

Figure 6.4.2 shows the physical diagram of the USB 2.0 section. U11 is the USB3320C, J3 is the Host USB interface, and J4 is the Slave USB interface. Jumpers J5 and J6 are used to select Host and Slave modes. The USB 2.0 pin assignments are shown in the table below :

Signal name	ZYNQ pin name	ZYNQ pin number	Remark
OTG_DATA4	PS_MIO28_501	C16	USB data bit 4
OTG_DIR	PS_MIO29_501	C13	USB data direction signal
OTG_STP	PS_MIO30_501	C15	USB stop signal
OTG_NXT	PS_MIO31_501	E16	USB next data signal
OTG_DATA0	PS_MIO32_501	A14	USB data bit 0
OTG_DATA1	PS_MIO33_501	D15	USB data bit 1
OTG_DATA2	PS_MIO34_501	A12	USB data bit 2
OTG_DATA3	PS_MIO35_501	F12	USB data bit 3
OTG_CLK	PS_MIO36_501	A11	USB clock signal
OTG_DATA5	PS_MIO37_501	A10	USB data bit 5
OTG_DATA6	PS_MIO38_501	E13	USB data bit 6
OTG_DATA7	PS_MIO39_501	C18	USB data bit 7
OTG_RESETN	PS_MIO 46_501	D16	USB reset signal

Table 6.4.2 - USB 2.0 Pin Assignment

6.5 USB to Serial Port

The AX7010 development board uses the Silicon Labs CP2102GM USB to UART chip. The USB interface is a Micro USB interface, which allows users to connect to a PC for serial communication using a Micro USB cable.

The UART's TX/RX signals are connected to the PS BANK501 of the ZYNQ EPP. Since the VCCMIO of this BANK is set to 1.8V, but the CP2102GM's data level is 3.3V, we use a TXS0102DCUR level converter chip for this connection. A schematic diagram of the CP2102GM and ZYNQ connection is shown in Figure 6.5.1 .

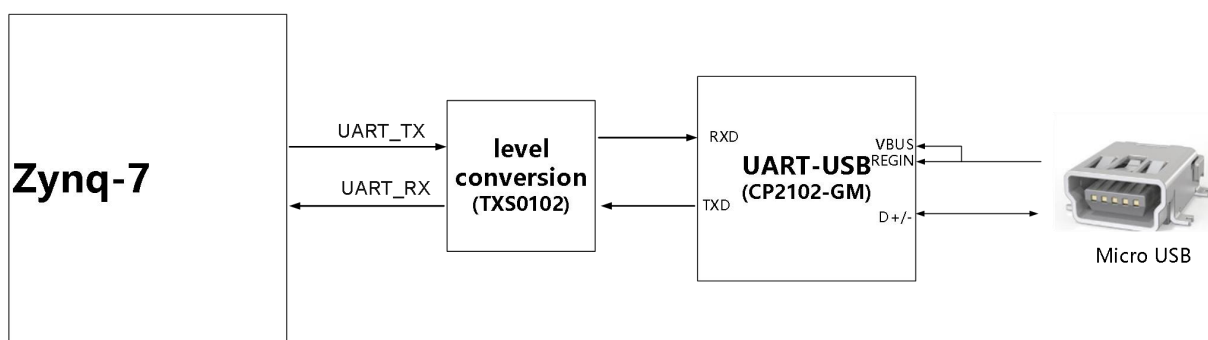


Figure 6.5.1 - Connection diagram of CP2102GM

ZYNQ serial port pin assignment:

Signal name	ZYNQ pin name	ZYNQ pin number	Remark
UART_TX	PS_MIO 48_501	B12	UART data output
UART_RX	PS_MIO 49_501	C12	UART data input

Table 6.5.1 - Serial Port Pin Assignment

Silicon Labs provides Virtual COM Port (VCP) drivers for the host PC. These drivers allow the CP2102GM USB-UART bridge device to appear as a COM port in communication applications such as TeraTerm or HyperTerminal. The VCP device driver must be installed before the host PC establishes communication with the AX7010 development board.

6.6 SD Card Slot

The AX7010 development board includes a MicroSD card interface, providing users with access to SD card storage for storing the ZYNQ chip's boot program, Linux operating system kernel, file system, and other user data files.

The SDIO signals are connected to the I/O signals of the ZYNQ's PS BANK501. Since the VCCMIO of this BANK is set to 1.8V, but the SD card's data level is 3.3V, we use a TXS02612 level converter for this connection. The schematic diagram of the Zynq7000 PS and SD card connector is shown in Figure 6.6.1.

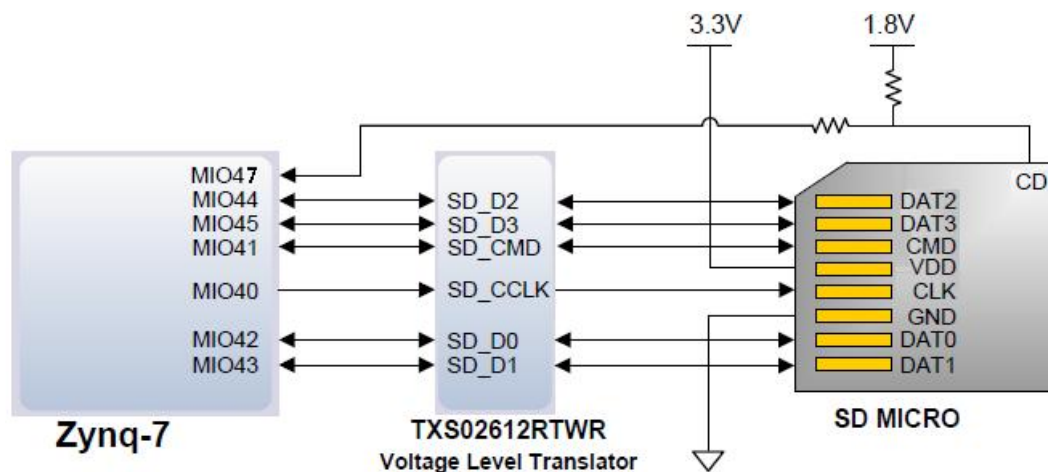


Figure 6.6.1 - SD Card Connection Diagram

SD card slot pin assignment

Signal name	ZYNQ pin name	ZYNQ pin number	Remark
SD_CLK	PS_MIO40	D14	SD clock signal
SD_CMD	PS_MIO41	C17	SD command signals
SD_D0	PS_MIO42	E12	SD Data0
SD_D1	PS_MIO43	A9	SD Data1
SD_D2	PS_MIO44	F13	SD Data2
SD_D3	PS_MIO45	B15	SD Data3
SD_CD	PS_MIO4 7	B14	SD card insertion signal

Table 6.6.1 - SD Card Slot Pin Assignment

6.7 PS PMOD Connector

The AX7010 development board has a 12-pin, 2.54mm pitch PMOD connector (J12) for connecting the PS BANK500's I/O to external modules or circuits. Because the BANK500's I/O is 3.3V standard, the signals from connected external devices and circuits also require a 3.3V level. The pin assignments for the PS PMOD connector are as follows .

PMOD pin	Signal name	ZYNQ pin name	ZYNQ pin number
PIN1	PMOD_IO0	PS_MIO11 _500	C6
PIN2	PMOD_IO 2	PS_MIO9 _500	B5

PIN3	PMOD_IO 3	PS_MIO 15_500	C8
PIN4	PMOD_IO 4	PS_MIO 7_500	D8
PIN5	GND	-	-
PIN6	+3.3V	-	-
PIN7	PMOD_IO 1	PS_MIO10_500	E9
PIN8	PMOD_IO 6	PS_MIO 8_500	D5
PIN9	PMOD_IO 7	PS_MIO 14_500	C5
PIN10	PMOD_IO 5	PS_MIO 12_500	D9
PIN11	GND	-	-
PIN12	+3.3V	-	-

Table 6.7.1 - Pin Assignment of PS PMOD Connector

6.8 User LED

On the AX7010 development board, two LEDs are connected to the BANK500 IO pin in the PS section. Users can use these LEDs to debug programs. When the BANK500 IO voltage is high, the LEDs are off; when the BANK500 IO voltage is low, the LEDs are on. A schematic diagram of the ZYNQ BANK500 IO pin and LED connection is shown in Figure 6.8.1.

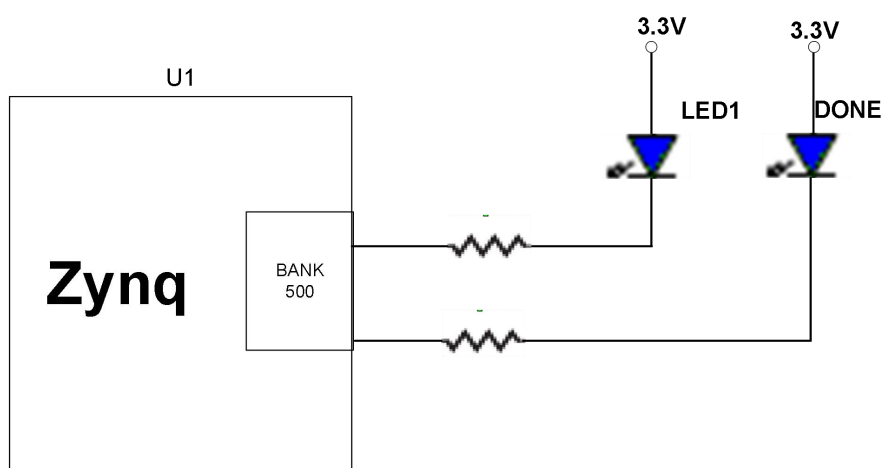


Figure 6.8.1 - Schematic diagram of Zynq-7000 and LED light connection

Pin assignment for PS LED lights

Signal name	ZYNQ pin name	ZYNQ pin number	Remark
MIO0_LED	PS_MIO 0_500	E6	PS LED1 light
MIO13_LED	PS_MIO 13_500	E8	PS LED2 lights

Table 6.8.1 - Pin Assignment for PS LEDs

6.9 User Button

On the AX7010 development board, two user buttons are connected to the BANK501 IO in the PS section. Users can use these buttons to test input signals and interrupt triggers. In the design, when a button is pressed, the signal voltage input to the ZYNQ BANK501 IO is low; when not pressed, the signal is high. A schematic diagram of the connection between the ZYNQ BANK501 IO and the buttons is shown in Figure 6.9.1 .

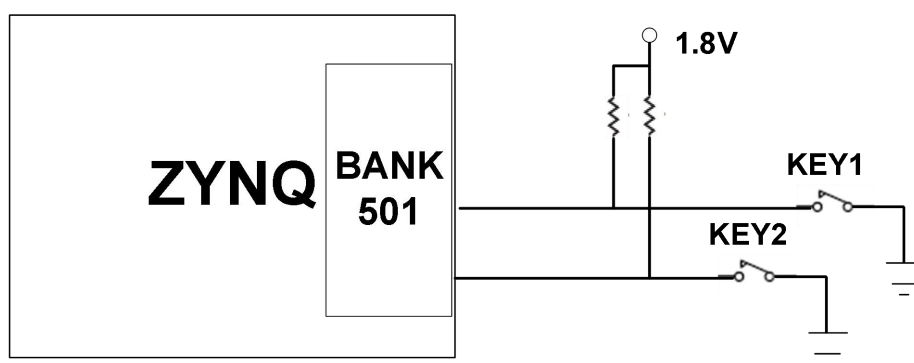


Figure 6.9.1 - Schematic diagram of Zynq-7000 and button connection

Key pin assignment

Signal name	ZYNQ pin name	ZYNQ pin number	Remark
MIO_KEY1	PS_MIO 50_501	B13	PS User Key KEY1
MIO_KEY 2	PS_MIO 51_501	B9	PS User Key KEY2

Table 6.9.1 - Key Pin Assignment

07 Peripherals at the PL end

Next, we will introduce the peripherals connected to the PL section (FPGA logic section).

7.1 HDMI Port

HDMI stands for High Definition Multimedia Video Output Interface. The AX7010 development

board connects directly to the differential signal and clock of the HDMI interface via the FPGA's differential I/O. Internally, the FPGA performs differential-to-parallel conversion of the HDMI signal before encoding and decoding, providing a DMI digital video input and output transmission solution that supports input and output up to 1080P@60Hz.

The HDMI signal is connected to BANK34 in the PL section of the ZYNQ. When the development board is used as an HDMI display device (HDMI IN), the HDMI signal is used as input and the HPD (hot plug detect) signal is used as output. When the development board is used as an HDMI host device (HDMI OUT), the opposite is true.

When the development board is used as an HDMI master device (HDMI OUT), it needs to provide a +5V power supply to the HDMI display device. Additionally, the HDMI master device reads the EDID device information of the HDMI display device via the IIC bus. The FPGA pin level is 3.3V, but the HDMI level is +5V; therefore, a level conversion chip GTL2002D is required for connection. The pin assignments for the HDMI interface are shown in the table below:

Signal name	ZYNQ pin name	ZYNQ number	pin	Remark
HDMI_CLK_P	IO_L13P_T2_MRCC_34	N18		HDMI clock signal positive
HDMI_CLK_N	IO_L13N_T2_MRCC_34	P19		HDMI clock signal negative
HDMI_D0_P	IO_L16P_T2_34	V20		HDMI data 0 positive
HDMI_D0_N	IO_L16 N_T2_34	W20		HDMI data 0 negative
HDMI_D1_P	IO_L15P_T2_DQS_34	T20		HDMI data 1 positive
HDMI_D1_N	IO_L15 N_T2_DQS_34	U20		HDMI data 1 negative
HDMI_D2_P	IO_L14P_T2_SRCC_34	N20		HDMI data 2 positive
HDMI_D2_N	IO_L14 N_T2_SRCC_34	P20		HDMI data 2 negative
HDMI_SCL	IO_L20N_T3_34	R18		HDMI IIC Clock
HDMI_SDA	IO_L19P_T2_34	R16		HDMI IIC data
HDMI_CEC	IO_L17P_T2_34	Y18		HDMI remote control signal
HDMI_HPD	IO_L17N_T2_34	Y19		HDMI hot-plug detection signal
HDMI_OUT_EN	IO_L18P_T2_34	V16		HDMI power output control

Table 7.1.1 - Pin Assignment for HDMI Interface

7.2 EEPROM

The AX7010 development board features an onboard EEPROM, model 24LC04, with a capacity of 4Kbit (2*256*8bit), consisting of two 256-byte blocks, communicating via the I2C bus. The onboard EEPROM is used to learn the I2C bus communication method. The EEPROM's I2C signal is connected to the BANK34 IO port on the ZYNQ PL pin. Figure 7.2.1 shows the EEPROM's schematic diagram .

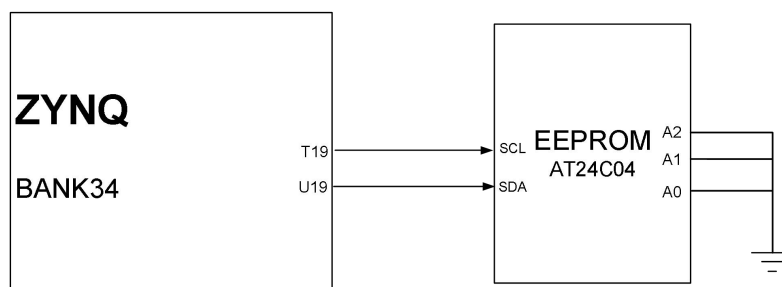


Figure 7.2.1 - Partial EEPROM Schematic Diagram

EEPROM pin assignment:

Signal name	ZYNQ pin name	ZYNQ pin number	Remark
EEPROM_I2C_SCL	IO_25_34	T19	IIC clock signal
EEPROM_I2C_S DA	IO_L12N_T1_MRCC_34	U19	IIC data signal

Table 7.2.1 - EEPROM Pin Assignment

7.3 Real-time clock

The development board has an onboard Real-Time Clock (RTC) chip, model DS1302, which provides a calendar up to 2099 , displaying the year, month, day, hour, minute, second, and day of the week. If the system requires timekeeping, the RTC needs to be integrated into the product. It requires an external 32.768kHz passive clock to provide a precise clock source to the chip, ensuring accurate clock information from the RTC. To ensure the RTC continues to operate even after power loss, a separate battery is typically used. Figure 7.3.1 shows BT1 as the battery holder. Inserting a **CR1220 3V coin cell battery allows the DS1302 to continue operating even** when the system is powered off . This ensures uninterrupted timekeeping regardless of power supply. The RTC's interface signals are connected to the BANK34 and BANK35 I/O ports on the ZYNQ PL connector. Figure 7.3.1 shows the schematic of the DS1302.

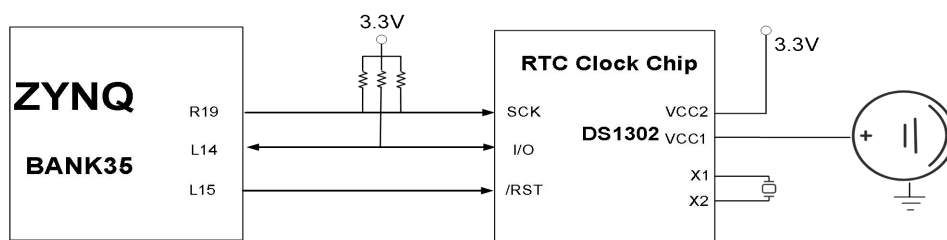


Figure 7.3.1 - DS1302 Schematic Diagram

DS1302 interface pin assignment:

Signal name	ZYNQ pin name	ZYNQ pin number	Remark
RTC_SCLK	IO_0_34	R19	RTC clock signal
RTC_RESET	IO_L22N_T3_AD7N_35	L15	RTC reset signal
RTC_DATA	IO_L22P_T3_AD7P_35	L14	RTC data signal

Table 7.3.1 - Clock Pin Assignment

7.4 Expansion Port J10

The expansion port J10 is a 40-pin 2.54mm dual-row connector, providing users with more peripherals and interfaces. Currently, ALINX offers modules including: **ADDA modules, LCD screen modules, Gigabit Ethernet modules, audio input/output modules, matrix keyboard modules, and 500W binocular vision camera modules**. The expansion port includes one 5V power supply, two 3.3V power supply lines, three ground lines, and 34 I/O ports. The I/O signals connect to the ZYNQ PL's BANK35 and BANK35, with a default voltage level of 3.3V. Some I/O signals on expansion port J10 can have their voltage levels changed by replacing the power supply chip (SPX3819M5-3-3) on the development board. **Do not connect directly to 5V devices to avoid damaging the FPGA. If you need to connect 5V devices, you need to use a level converter chip.**

A 33-ohm resistor array is connected in series between the expansion port and the FPGA to protect the FPGA from damage caused by excessive external voltage or current. The P and N traces on the PCB design use differential routing, controlling the differential impedance to 100 ohms. J10 expansion port pin assignment.

J10 pin	Signal name	ZYNQ pin name	ZYNQ pin number
PIN1	GND	-	-
PIN2	+5V	-	-
PIN3	EX_IO1_1N	IO_L22 N_T3_34	W19

PIN4	EX_IO1_1 P	IO_L22P_T3_34	W18
PIN5	EX_IO1_2 N	IO_L6N_T0_34	R14
PIN6	EX_IO1_2P	IO_L6P_T0_34	P14
PIN7	EX_IO1_3 N	IO_L7 N_T1_34	Y17
PIN8	EX_IO1_3P	IO_L7P_T1_34	Y16
PIN9	EX_IO1_4 N	IO_L10 N_T1_34	W15
PIN10	EX_IO1_4P	IO_L10P_T1_34	V15
PIN11	EX_IO1_5 N	IO_L8 N_T1_34	Y14
PIN12	EX_IO1_5P	IO_L8P_T1_34	W14
PIN13	EX_IO1_6 N	IO_L23N_T3_34	P18
PIN14	EX_IO1_6P	IO_L23 P_T3_34	N17
PIN15	EX_IO1_7 N	IO_L11 N_T1_34	U15
PIN16	EX_IO1_7P	IO_L11P_T1_34	U14
PIN17	EX_IO1_8 N	IO_L24 N_T3_34	P16
PIN18	EX_IO1_8P	IO_L24P_T3_34	P15
PIN19	EX_IO1_9 N	IO_L9N_T1_34	U17
PIN20	EX_IO1_9P	IO_L9P_T1_34	T16
PIN21	EX_IO1_10 N	IO_L21_N_T3_34	V18
PIN22	EX_IO1_10P	IO_L21_P_T3_34	V17
PIN23	EX_IO1_11 N	IO_L5 N_T0_34	T15
PIN24	EX_IO1_11P	IO_L5P_T0_34	T14
PIN25	EX_IO1_12 N	IO_L3 N_T0_34	V13
PIN26	EX_IO1_12P	IO_L3P_T0_34	U13
PIN27	EX_IO1_13 N	IO_L4 N_T0_34	W13
PIN28	EX_IO1_13P	IO_L4P_T0_34	V12
PIN29	EX_IO1_14 N	IO_L2 N_T0_34	U12
PIN30	EX_IO1_14P	IO_L2P_T0_34	T12

PIN31	EX_IO1_15 N	IO_L1 N_T0_34	T10
PIN32	EX_IO1_15P	IO_L1P_T0_34	T11
PIN33	EX_IO1_16 N	IO_L2 N_T0_35	A20
PIN34	EX_IO1_16P	IO_L2P_T0_35	B19
PIN35	EX_IO1_17 N	IO_L1 N_T0_35	B20
PIN36	EX_IO1_17P	IO_L1P_T0_35	C20
PIN37	GND	-	-
PIN38	GND	-	-
PIN39	+3.3V	-	-
PIN40	+3.3V	-	-

Table 7.4.1 - J10 Expansion Port Pin Assignment

7.5 Expansion Port J11

The expansion port J11 is also a 40-pin 2.54mm dual-row connector, allowing users to expand with more peripherals and interfaces. Currently, ALINX provides modules such as: **ADDA modules**, **LCD screen modules**, **Gigabit Ethernet modules**, **audio input/output modules**, **matrix keyboard modules**, and **500W binocular vision camera modules**. The expansion port includes one 5V power supply, two 3.3V power supply, three ground lines, and 34 I/O ports. The I/O signals are connected to the ZYNQ PL's BANK35, with a default voltage level of 3.3V. All I/O signals on expansion port J11 can be changed by replacing the power supply chip (SPX3819M5-3-3) on the development board. **Do not connect directly to 5V devices to avoid damaging the FPGA. If you need to connect 5V devices, you need to use a level converter chip.**

A 33-ohm resistor array is connected in series between the expansion port and the FPGA to protect the FPGA from damage caused by excessive external voltage or current. The P and N traces on the PCB are routed differentially, with the differential impedance controlled at 100 ohms. J11 expansion port pin assignment.

J11 pin	Signal name	ZYNQ pin name	ZYNQ pin number
PIN1	GND	-	-
PIN2	+5V	-	-
PIN3	EX_IO 2_1N	IO_L6 N_T0_35	F17
PIN4	EX_IO 2_1 P	IO_L6P_T0_35	F16

PIN5	EX_IO 2_2 N	IO_L15 N_T2_35	F20
PIN6	EX_IO 2_2P	IO_L15P_T2_35	F19
PIN7	EX_IO 2_3 N	IO_L18 N_T2_35	G20
PIN8	EX_IO 2_3P	IO_L18P_T2_35	G19
PIN9	EX_IO 2_4 N	IO_L14N_T2_35	H18
PIN10	EX_IO 2_4P	IO_L14 P_T2_35	J18
PIN11	EX_IO 2_5 N	IO_L9 N_T1_35	L20
PIN12	EX_IO 2_5P	IO_L9P_T1_35	L19
PIN13	EX_IO 2_6 N	IO_L7 N_T1_35	M20
PIN14	EX_IO 2_6P	IO_L7P_T1_35	M19
PIN15	EX_IO 2_7 N	IO_L12 N_T1_35	K18
PIN16	EX_IO 2_7P	IO_L12P_T1_35	K17
PIN17	EX_IO 2_8 N	IO_L10 N_T1_35	J19
PIN18	EX_IO 2_8P	IO_L10P_T1_35	K19
PIN19	EX_IO 2_9 N	IO_L17 N_T2_35	H20
PIN20	EX_IO 2_9P	IO_L17P_T2_35	J20
PIN21	EX_IO 2_10 N	IO_L11 N_T1_35	L17
PIN22	EX_IO 2_10P	IO_L11P_T1_35	L16
PIN23	EX_IO 2_11 N	IO_L8 N_T1_35	M18
PIN24	EX_IO 2_11P	IO_L8P_T1_35	M17
PIN25	EX_IO 2_12 N	IO_L4 N_T0_35	D20
PIN26	EX_IO 2_12P	IO_L4P_T0_35	D19
PIN27	EX_IO 2_13 N	IO_L5 N_T0_35	E19
PIN28	EX_IO 2_13P	IO_L5P_T0_35	E18
PIN29	EX_IO 2_14 N	IO_L16 N_T2_35	G18
PIN30	EX_IO 2_14P	IO_L16P_T2_35	G17
PIN31	EX_IO 2_15 N	IO_L13 N_T2_35	H17

PIN32	EX_IO 2_15P	IO_L13P_T2_35	H16
PIN33	EX_IO 2_16 N	IO_L19 N_T3_35	G15
PIN34	EX_IO 2_16P	IO_L19P_T3_35	H15
PIN35	EX_IO 2_17 N	IO_L20 N_T3_35	J14
PIN36	EX_IO 2_17P	IO_L20P_T3_35	K14
PIN37	GND	-	-
PIN38	GND	-	-
PIN39	+3.3V	-	-
PIN40	+3.3V	-	-

Table 7.5.1 - J11 Expansion Port Pin Assignment

7.6 User LED

The PL section of the AX7010 development board connects to four LEDs. The schematic diagram of the four user LEDs is shown in Figure 7.6.1 . The LED signals are connected to the IO pins of the BANK35 pin in the PL section. When the IO pin output of the BANK35 pin is logic 0, the LED is lit; when the output is logic 1, the LED is turned off.

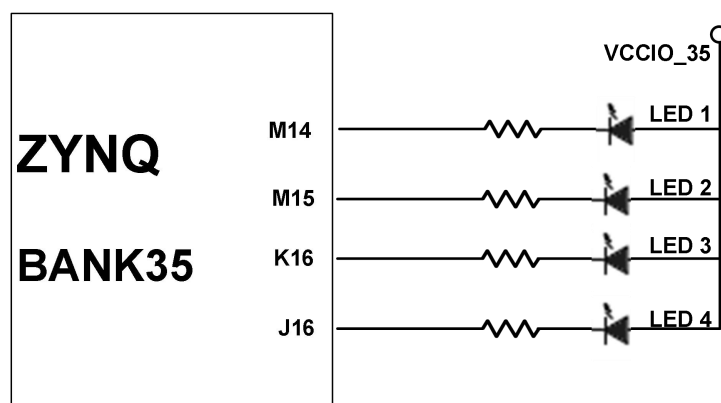


Figure 7.6.1 - PL User LED Schematic Diagram

PL user LED pin assignment:

Signal name	ZYNQ pin name	ZYNQ pin number	Remark
LED1	IO_L23P_T3_35	M14	PL user LED1 light
LED2	IO_L23N_T3_35	M15	PL user LED2 lights

LED3	IO_L24P_T3_35	K16	PL user LED 3 lights
LED4	IO_L24N_T3_35	J16	PL user LED 4 lights

Table 7.6.1-PL User LED Pin Assignment

7.7 User Buttons

The PL section of the AX7010 development board has four user buttons (KEY1~KEY4). The button signals are connected to the I/O pins of ZYNQ's BANK34 and BANK35. **All buttons are active low** ; the signal is high when not pressed and low when pressed. The schematic diagram of the four user buttons is shown in Figure 7.7.1 .

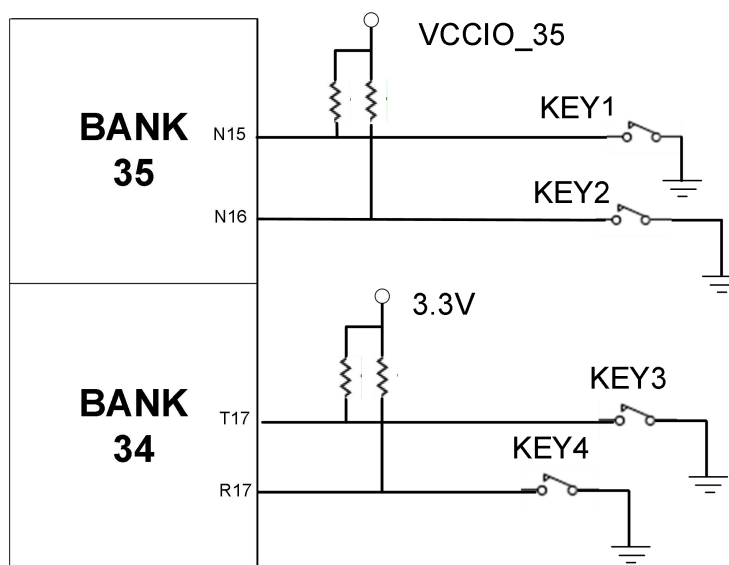


Figure 7.7.1 - Schematic diagram of user button operation

Key pin assignment:

Signal name	ZYNQ pin name	ZYNQ pin number	Remark
KEY1	IO_L21P_T3_35	N15	PL User Button 1
KEY2	IO_L21N_T3_35	N16	PL user button 2
KEY3	IO_L20P_T3_34	T17	PL user button 3
KEY4	IO_L19N_T3_34	R17	PL user button 4

Table 7.7.1 - Key Pin Assignment



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